

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First/ Second Semester of B. Tech Theory Examination (CL/ME/EE/CE/IT/EC)****April – May 2018****CL 103 / CL103.01 Mechanics of Solids****Date: 11.05.2018, Friday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

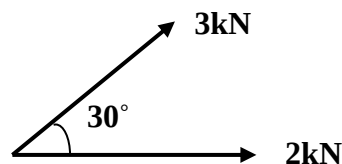
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat sketch wherever required.
4. **Use of scientific calculator is allowed.**

SECTION – I**Q - 1 Do as directed. (Attempt any seven) [07]**

- i) N/mm^2 is the unit of _____.
(a) Stress (b) Force (c) Weight
- ii) A _____ quantity is one that has magnitude as well as direction.
(a) Vector (b) Volume (c) Area
- iii) Enumerate the characteristics of a force.
- iv) A body which deforms under the action of applied forces is called the _____.
(a) Space (b) Deformable body (c) Static
- v) State Newton's third law.
- vi) _____ is the condition for perfect truss.
(a) $m \neq 2j-r$ (b) $m = 2j-r$ (c) $m > 2j-r$
- vii) "If three forces acting at a point are in equilibrium, then each force will be proportional to the sine of angle between other two forces". This is the statement of _____.
(a) Parallelogram law of forces (b) Lami's theorem (c) Varignon's theorem
- viii) Law of moments is also called the Varignon's theorem. True or false?

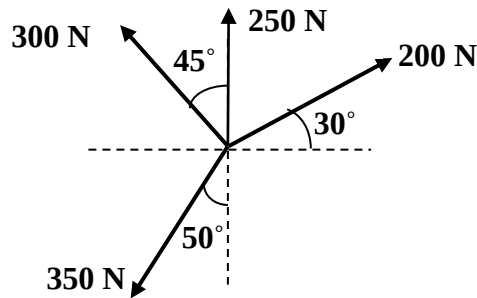
Q - 2 (a) Enlist the types of force system and explain any one in detail. [04]**OR**

Q - 2 (a) Two forces of magnitude 2 kN and 3 kN are acting at a point as shown in Figure 1. [04]
Calculate the resultant of these concurrent force system.

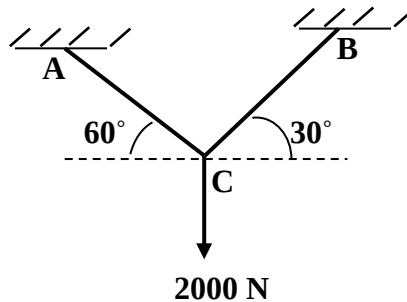
**Figure 1 [Q-2 (a) OR]**

Q - 2 (b) Attempt any two.**[10]**

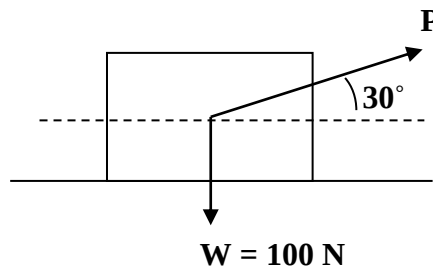
- i) What is couple? Enumerate its various characteristics.
- ii) Calculate the resultant of coplanar concurrent force system shown in **Figure 2**.

**Figure 2 [Q-2 (b) (ii)]**

- iii) State and explain Parallelogram law of forces.
- iv) A weight of 2000 N is suspended from rope AC and BC as shown in **Figure 3**. Calculate the tension in each rope.

**Figure 3 [Q-2 (b) (iv)]**

- Q - 3 (a)** A body of weight 100 N rests on a rough horizontal plane and is acted upon by a force **[07]** applied at an angle of 30° to the horizontal. Determine the force required to just cause the body to slide over the surface in the direction of applied force. Take coefficient of static friction as 0.3. Refer **Figure 4**.

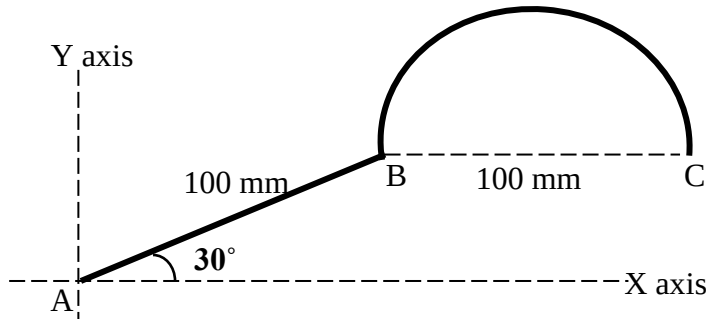
**Figure 4 [Q-3 (a)]****OR**

- Q - 3 (a)** i) Explain with the help of a neat diagram, the concept of limiting friction. **[03]**
- ii) Friction is necessary evil. Comment on the validity of the statement. **[04]**

- Q - 3 (b)** i) Differentiate between center of gravity and centroid. [03]
 ii) Give the position of centroid of the following standard sections: [04]
 Rectangle, Triangle, Solid circle, Semicircle

OR

- Q - 3 (b)** Find the centroid of the bent weir shown in **Figure 5**. Consider length of AB is 100 mm [07]
 and diameter of semicircle BC is 100 mm.

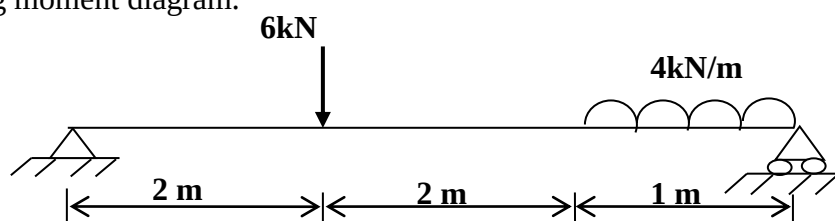
**Figure 5 [Q-3 (b) OR]****SECTION – II**

- Q - 4** Do as directed. [07]

- Ratio of ultimate load to cross sectional area is called _____.
 (a) Ultimate stress (b) Impact load (c) FOS
- The property of the material to be drawn into thin sheet is known as _____.
 (a) Malleability (b) Tension (c) Torque
- Direct stress is the ratio of axial force to cross sectional area. True or false?
- S.I. unit of Modulus of elasticity is _____.
- _____ is a measure of the resistance to surface indentation or abrasion.
 (a) Ductility (b) Hardness (c) Shear
- _____ is the machine used for conducting Impact Test.
 (a) Izod impact test (b) UTM (c) Torsion test
- Define factor of safety.

- Q - 5** Attempt any two. [14]

- Enlist and explain the types of support and types of load with neat sketch.
- Analyse the simply supported beam of span 5 m shown in **Figure 6**. Also draw shear force and bending moment diagram.

**Figure 6 [Q-5 (b)]**

- (c) Analyse the cantilever beam of span 4 m shown in **Figure 7**. Also draw shear force and bending moment diagram.

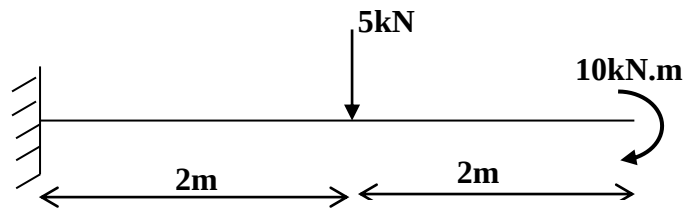


Figure7 [Q-5 (c)]

- Q – 6 (a)** A steel wire of diameter 20 mm and length 2.5 m is subjected to an axial pull of 125 kN. Find stress developed in the rod. **[04]**

- Q – 6 (b)** **Attempt any two** **[10]**

- A 25 mm diameter steel rod of length 3 m is subjected to an axial pull of 30 kN. If modulus of elasticity is 200 GPa and Poisson's ratio 0.25, find the change in length, change in diameter and change in volume.
- Define the terms: longitudinal strain, lateral strain, Poisson's ratio, volumetric strain and axial stress
- Find the total deformation of the rod ABCD shown in **Figure 8**. Take $E = 200$ GPa.

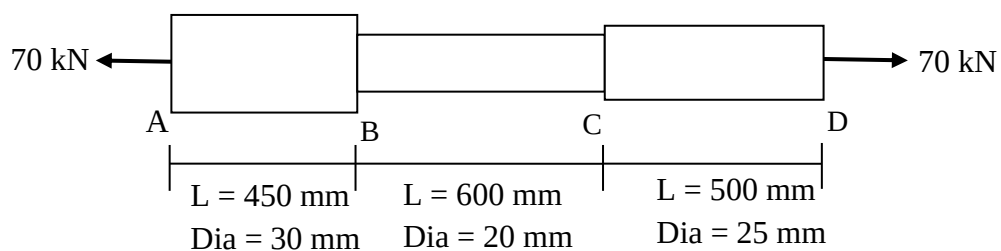


Figure8 [Q-6 (b) (iii)]

- State Hooke's law and explain it in brief.
